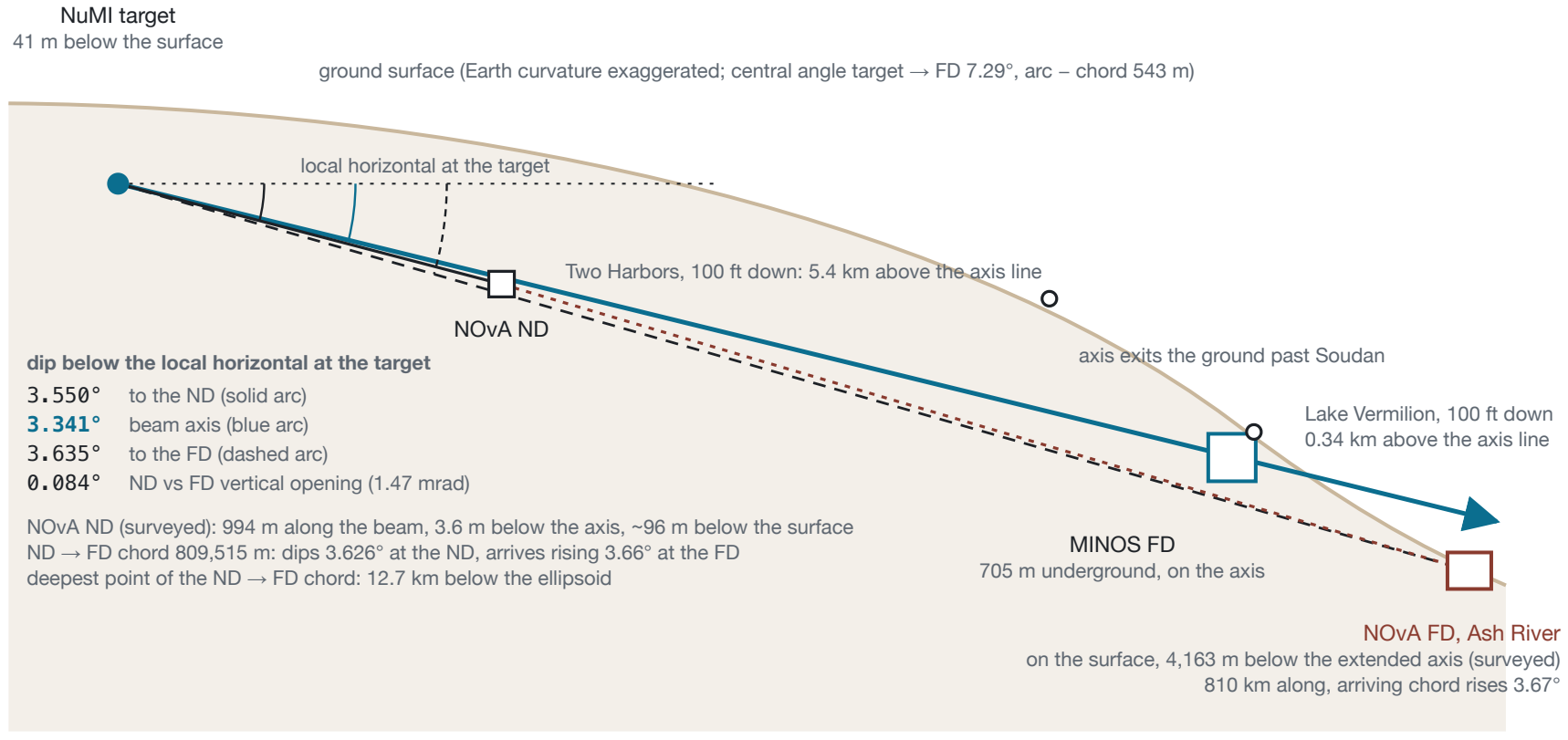
NuMI target
41 m below the surface
ground surface (Earth curvature exaggerated; central angle target → FD 7.29°, arc − chord 543 m)
local horizontal at the target
Two Harbors, 100 ft down: 5.4 km above the axis line
NOvA ND
axis exits the ground past Soudan
Lake Vermilion, 100 ft down
0.34 km above the axis line
dip below the local horizontal at the target
3.550° to the ND (solid arc)
3.341° beam axis (blue arc)
3.635° to the FD (dashed arc)
0.084° ND vs FD vertical opening (1.47 mrad)
NOvA ND (surveyed): 994 m along the beam, 3.6 m below the axis, ~96 m below the surface
ND → FD chord 809,515 m: dips 3.626° at the ND, arrives rising 3.66° at the FD
deepest point of the ND → FD chord: 12.7 km below the ellipsoid
MINOS FD
705 m underground, on the axis
NOvA FD, Ash River
on the surface, 4,163 m below the extended axis (surveyed)
810 km along, arriving chord rises 3.67°